

Section 100 — Grandmaster Finale

How to use the finale

This section is an examination and synthesis bank, not a catalogue of new named algorithms. Its sixty-five original problems reuse material from the course under less explicit modeling. Attempt each problem before opening its editorial.

For every attempt, write five items:

1. the literal slow model;
2. the bottleneck preventing it from meeting the limits;
3. the lifetime, separator, algebraic identity, or monotonicity that removes that bottleneck;
4. the compressed state and the information it deliberately forgets;
5. a tiny independent oracle.

Difficulty is progressive inside each round and across rounds. Problems 01–10 emphasize histories and tree separators. Problems 11–20 cover optimization, flow, matching, and cuts. Problems 21–30 cover strings and sequences. Problems 31–40 cover algebra and number theory. Problems 41–49 cover geometry, grids, and subset search. Problems 50–65 are the grandmaster gauntlet: each hides one decisive idea behind an ordinary-looking statement.

Round I contract

The first ten published problems share one question: what information survives when time branches or a tree is compressed?

Do not infer the solution merely from the problem order. Persistence, rollback, offline divide and conquer, dynamic trees, centroids, virtual trees, Euler intervals, HLD, and reconstruction trees appear in different roles. The exercise is to recognize which boundary the future can still cross.

Submission checklist

- Prove that every queried object maps to exactly one compressed interval, version, separator, or state.
- Prove that no object is counted in two branches or decomposition levels.
- Audit strict versus non-strict comparisons and all tie rules.
- Use wide arithmetic before modular reduction.
- Test a chain, a star, one element, all equal keys, and the smallest legal history.
- Compare against the supplied independent random oracle before optimizing.

Round II contract

Problems 11–20 ask the same modeling question in optimization form: what is the smallest network, dual variable, exchange graph, or convex boundary that still certifies the optimum? Lower bounds become vertex imbalances; exact counts become a Lagrange multiplier plus a tie interval; laminar choices become shared tree capacity. Write down that reduction before choosing an algorithm.

Exercises

All six rounds contain Problems 01–65. Every published package has a statement, independent oracle, two references, proof notes, and rendering checks.

Round III contract

String structures become useful only after a substring is translated into a state, suffix interval, or first-occurrence certificate. Problems 21–30 ask which numeric query remains after that translation. Keep equality boundaries, suffix-link propagation limits, and the difference between occurrences and distinct strings explicit.

Round IV contract

Problems 31–40 are algebra and number theory without labels. Before computing, name the structure: an exponential formula, a hidden polynomial, a functional equation, a vector space over $\text{GF}(2)$, a group order, a derivation. Then state the field or ring you work in, and every reduction ($n \bmod p$, $k \bmod p$, $n \bmod (p-1)$) together with the reason it is valid.

Round V contract

Problems 41–49 mix exact geometry with grid and subset search. Use integer predicates wherever the input is integral, and floating point only for the final lengths and areas. For each problem, identify the boundary object that carries the answer (uncovered arcs, the horizon of a hull, the first row of presses, a half of the vertex set) and prove that nothing else matters.

Round VI contract

Problems 50–65 do not announce their technique, and several look like routine exercises until the limits are read. Before coding, find the object that makes the problem small: a hidden linear recurrence, a dominator, the last occurrence of a substring, the newest spanning forest, a run of a string, the upper envelope of all paths. Then prove that it keeps the answer unchanged. Each package ships a brute-force oracle; use it before tuning for speed.